INNOMATICS®
RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Employee Management System

Relational Database Design & Analytics using MySQL

Presented by:

G. Akhil Yadav
M. Swamy Yadav
N. Sai Venkatesh Kumar

https://www.linkedin.com/in/akhilyadav25/
https://github.com/akhilyadav2502-stack

About Data Analytics

Data Analytics

Translating raw HR, payroll and leave data into actionable business insights using structured query logic.

Why Data Analytics?

Data bridges finance and strategy. Evidence-based decisions drive better outcomes in any business domain.

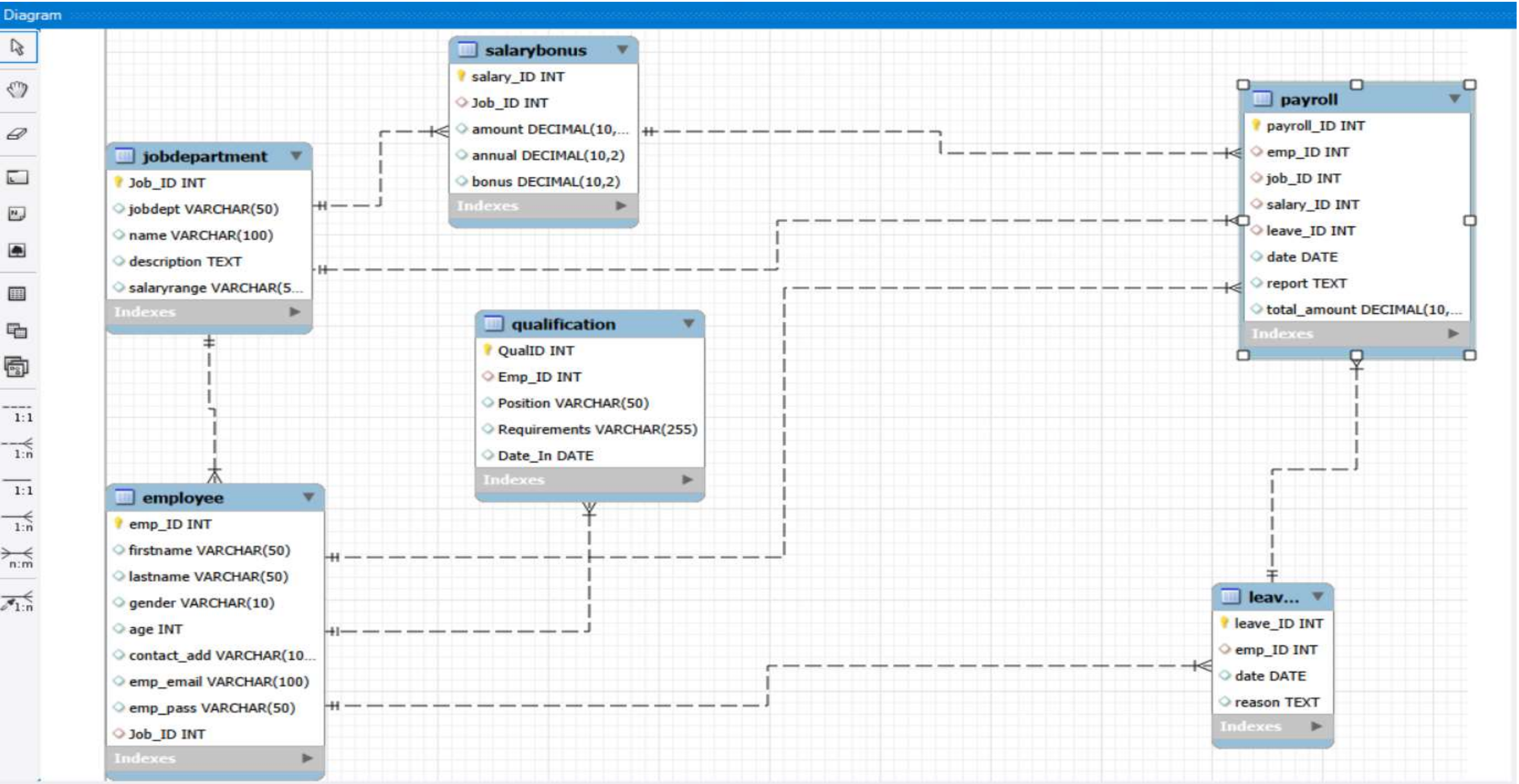
Project Scope

60 employees · 6 normalized tables · 21 analytical queries across HR, payroll, qualifications and leave.

AGENDA

- Objective of the Project
- Employee Insights
- Job Role & Department Analysis
- Qualification & Skills Analysis
- Leave & Absence Patterns
- Payroll & Compensation Analysis
- Challenges, Conclusion

ER-DIAGRAM / Reverse - Engineering



Tables

Job Department table

Job_ID	jobdept	name	description	salaryrange
1	Operations	Operations Engineer	Responsible for engineer duties in the operation...	$70889 - $91502
2	Finance	Finance Consultant	Responsible for consultant duties in the finance ...	$45757 - $102274
3	IT	IT Coordinator	Responsible for coordinator duties in the IT dep...	$32637 - $118326
4	Marketing	Marketing Engineer	Responsible for engineer duties in the marketin...	$48936 - $145464
5	Finance	Finance Specialist	Responsible for specialist duties in the finance d...	$67562 - $99403
6	Engineering	Engineering Executive	Responsible for executive duties in the enginee...	$66000 - $125000
7	Sales	Sales Manager	Responsible for managing the sales department.	$75000 - $130000
8	IT	IT Analyst	Responsible for analyzing IT systems in the IT d...	$48000 - $92000
9	HR	HR Manager	Responsible for managing the human resources ...	$68000 - $105000
10	Operations	Operations Specialist	Responsible for specialist duties in the operatio...	$54000 - $90000
11	Marketing	Marketing Coordinator	Responsible for coordinating marketing campaig...	$55000 - $95000
12	Legal	Legal Advisor	Responsible for advising on legal matters in the ...	$95000 - $150000
13	Finance	Financial Analyst	Responsible for analyzing financial data and tre...	$58000 - $120000

Salary table

salary_ID	Job_ID	amount	annual	bonus
1	1	45000.00	540000.00	5000.00
2	2	55000.00	660000.00	7000.00
3	3	40000.00	480000.00	4000.00
4	4	48000.00	576000.00	6000.00
5	5	47000.00	564000.00	4500.00
6	6	65000.00	780000.00	10000.00
7	7	80000.00	960000.00	12000.00
8	8	42000.00	504000.00	3500.00
9	9	55000.00	660000.00	7000.00
10	10	48000.00	576000.00	5500.00
11	11	52000.00	624000.00	6000.00
12	12	90000.00	1080000.00	15000.00
13	13	60000.00	720000.00	8000.00

Employees table

emp_ID	firstname	lastname	gender	age	contact_add	emp_email	emp_pass	Job_ID
1	John	Doe	M	30	123 Elm St, NY	john.doe@example.com	password123	1
2	Jane	Smith	F	28	456 Maple Ave, CA	jane.smith@example.com	password123	2
3	Alice	Brown	F	35	789 Oak St, TX	alice.brown@example.com	password123	3
4	Bob	Johnson	M	40	101 Pine St, FL	bob.johnson@example.com	password123	4
5	Charlie	Williams	M	33	202 Birch Rd, IL	charlie.williams@example.com	password123	5
6	Dave	Miller	M	29	303 Cedar Blvd, NY	dave.miller@example.com	password123	6
7	Eve	Davis	F	25	404 Maple Dr, TX	eve.davis@example.com	password123	7
8	Frank	Wilson	M	38	505 Pine Ave, FL	frank.wilson@example.com	password123	8
9	Grace	Moore	F	45	606 Oak Blvd, CA	grace.moore@example.com	password123	9
10	Harry	Taylor	M	50	707 Elm Rd, NY	harry.taylor@example.com	password123	10
11	Irene	Anderson	F	32	808 Maple St, FL	irene.anderson@example.com	password123	11
12	Jack	Thomas	M	34	909 Pine Rd, TX	jack.thomas@example.com	password123	12
13	Karen	Jackson	F	30	1010 Oak Dr, NY	karen.jackson@example.com	password123	13

Qualification Table

QualID	Emp_ID	Position	Requirements	Date_In
1	1	HR Assistant	BBA Communication Skills	2019-12-06
2	2	HR Manager	MBA 5 Yrs Exp	2018-03-20
3	3	IT Support	B.Tech (IT) Troubleshooting	2020-01-15
4	4	Software Engineer	B.Tech (CS) Java/Python	2021-04-01
5	5	Data Analyst	B.Sc Stats Excel SQL	2020-08-11
6	6	Sr. Data Analyst	M.Sc Stats Power BI	2017-11-20
7	7	Marketing Executive	BBA Google Ads	2021-05-10
8	8	Marketing Manager	MBA Marketing 5 Yrs Exp	2018-07-14
9	9	Finance Analyst	B.Com Excel SAP	2020-02-19
10	10	Sr. Finance Officer	M.Com Chartered Accountant	2016-12-01
11	11	Legal Advisor	LLB Corporate Law	2015-10-09
12	12	Office Admin	B.A MS Office	2019-05-03
13	13	HR Executive	MBA HR Recruitment Skills	2018-08-22

Leaves Table

leave_ID	emp_ID	date	reason
1	1	2024-01-15	Medical Leave
2	2	2024-10-02	Personal Work
3	3	2024-05-03	Sick Leave
4	4	2024-01-25	Family Emergency
5	5	2024-10-03	Travel
6	6	2024-02-15	Medical Checkup
7	7	2024-01-04	Festival
8	8	2024-10-04	Vacation
9	9	2024-02-20	Illness
10	10	2024-05-01	Personal Work
11	11	2024-03-20	Medical Emergency
12	12	2024-01-30	Marriage
13	13	2024-03-25	Maternity Leave

Payroll table

payroll_ID	emp_ID	job_ID	salary_ID	leave_ID	date	report	total_amount
1	1	1	1	1	2024-04-30	April Payroll	40500.00
2	2	2	2	2	2024-04-30	April Payroll	45500.00
3	3	3	3	3	2024-04-30	April Payroll	39500.00
4	4	4	4	4	2024-04-30	April Payroll	43500.00
5	5	5	5	5	2024-04-30	April Payroll	38500.00
6	6	6	6	6	2024-04-30	April Payroll	42500.00
7	7	7	7	7	2024-04-30	April Payroll	44500.00
8	8	8	8	8	2024-04-30	April Payroll	45500.00
9	9	9	9	9	2024-04-30	April Payroll	40500.00
10	10	10	10	10	2024-04-30	April Payroll	38500.00
11	11	11	11	11	2024-04-30	April Payroll	49500.00
12	12	12	12	12	2024-04-30	April Payroll	51500.00
13	13	13	13	13	2024-04-30	April Payroll	50500.00

Project Objective

About the Project

- Build a relational Employee Management System using MySQL
- Design normalized tables: Employee, JobDepartment, SalaryBonus, Qualification, Leaves, Payroll
- Write SQL queries to extract HR, payroll, and performance insights
- 60 employees across 10+ departments with 21 analytical queries

Database Tables

- Employee → JobDepartment → SalaryBonus
- Employee → Leaves → Payroll
- Employee → Qualification

Key Constraints Used

- PRIMARY KEY on every table
- FOREIGN KEY with ON DELETE CASCADE / SET NULL
- UNIQUE constraint on emp_email
- DATE fields in YYYY-MM-DD format

Challenges

- Defining correct FK relationships
- Cascading updates & deletes
- Complex multi-table JOINs
- Preventing duplicate emails

Employee Insights

Overview

This section explores the employee table to understand headcount, salary distribution, and expenditure across the company.

Q1: How many unique employees are in the system?

```
SELECT DISTINCT * FROM employee;
```

Returns all 60 unique employee records. The DISTINCT keyword ensures no duplicate rows appear.

Q2: Which department has the highest number of employees?

```
SELECT jobdept, COUNT(Job_id) AS No_of_Employees
FROM jobdepartment
GROUP BY jobdept ORDER BY No_of_Employees DESC LIMIT 1;
```

Groups jobs by department, counts roles, and returns the top department by headcount.

Result Preview

Total Employees: 60

Top Department by Count:

Finance — 9 Employees

Key SQL Concepts Used:

- **DISTINCT — removes duplicate rows**
- **COUNT() — counts matching rows**
- **GROUP BY — aggregates per department**
- **ORDER BY DESC — sorts highest first**
- **LIMIT 1 — returns only top result**

Employee Insights (Contd.)

Q3: What is the average salary per department?

```
SELECT jd.jobdept, AVG(sb.amount) AS avg_salary
FROM JobDepartment jd
JOIN SalaryBonus sb ON jd.Job_ID = sb.Job_ID
GROUP BY jd.jobdept;
```

Joins JobDepartment with SalaryBonus and computes average monthly salary per department.

Q4: Who are the top 5 highest-paid employees?

```
SELECT e.emp_ID, e.firstname, e.lastname, sb.amount
FROM Employee e JOIN SalaryBonus sb ON e.Job_ID =
sb.Job_ID
ORDER BY sb.amount DESC LIMIT 5;
```

Joins Employee and SalaryBonus tables, sorts by salary descending, returns top 5.

Q5: What is the total salary expenditure across the company?

```
SELECT SUM(total_amount) AS total_salary_expenditure FROM
Payroll;
```

Aggregates the total_amount column across all payroll records.

Average Salary per Department

jobdept	avg_salary
Operations	68750.000000
Finance	72333.333333
IT	70888.888889
Marketing	65625.000000
Engineering	81142.857143
Sales	75428.571429
HR	62571.428571
Legal	84600.000000

Top 5 Paid Employees:

emp_ID	firstname	lastname	salary
37	Ingrid	Adams	170000.00
38	John	Baker	160000.00
35	Grake	Moor	150000.00
36	Hank	Wilson	150000.00
39	Kelly	Cooper	140000.00

Total Payroll Expenditure:

total_salary_expenditure
2778000.00

Job Role & Department Analysis

Q6: How many different job roles exist in each department?

```
SELECT jobdept, COUNT(DISTINCT Job_ID) AS total_roles
FROM JobDepartment GROUP BY jobdept;
```

Counts unique Job_IDs per department. Finance, IT, Engineering each have 8+ distinct roles.

Q7: What is the average salary range per department?

```
SELECT jd.jobdept, AVG(sb.amount) AS avg_salary
FROM JobDepartment jd JOIN SalaryBonus sb ON jd.Job_ID =
sb.Job_ID
GROUP BY jd.jobdept;
```

Confirms which departments offer the highest average pay.

Q8: Which job role offers the highest salary?

```
SELECT jd.Job_ID, jd.jobdept, MAX(sb.amount) AS
highest_salary
FROM JobDepartment jd JOIN SalaryBonus sb ON jd.Job_ID =
sb.Job_ID
GROUP BY jd.Job_ID, jd.jobdept ORDER BY highest_salary
DESC LIMIT 1;
```

Uses MAX() aggregation. Result: Finance Director at $170,000/month.

Departments And Job roles:

jobdept	total_roles
Engineering	7
Finance	9
HR	7
IT	9
Legal	5
Marketing	8
Operations	8
Sales	7

Department And Avg_Salary:

jobdept	avg_salary
Operations	68750.000000
Finance	72333.333333
IT	70888.888889
Marketing	65625.000000
Engineering	81142.857143
Sales	75428.571429
HR	62571.428571
Legal	84600.000000

Highest salary of To Particular Role:

Job_ID	jobdept	highest_salary
37	Finance	170000.00

Job Role & Department Analysis

Q9:Which Departments have the Highest Total Salary Allocation

```
SELECT jd.jobdept, SUM(sb.amount) As total_salary
FROM Employee e
JOIN JobDepartment jd ON e.Job_ID = jd.Job_ID
JOIN SalaryBonus sb ON jd.Job_ID = sb.Job_ID
GROUP BY jd.jobdept ORDER BY total_salary DESC
LIMIT 3;
```

Triple JOIN: links Employee → JobDepartment → SalaryBonus. SUM aggregates payroll per dept.

jobdept	total_salary
Finance	651000.00
IT	638000.00
Engineering	568000.00

Qualification & Skills Analysis

Q10: How many employees have at least one qualification?

```
SELECT COUNT(DISTINCT Emp_ID) AS employees_with_qualification
FROM Qualification;
```

Counts distinct employees in the Qualification table. Result: All 60 employees have at least one qualification listed.

Q11: Which positions require the most qualifications?

```
SELECT Position, COUNT(*) AS total_qualifications
FROM Qualification
GROUP BY Position ORDER BY total_qualifications DESC LIMIT 3;
```

Groups qualifications by Position and counts entries. Identifies top 3 most requirement-heavy roles.

Q12: Which employees have the highest number of qualifications?

```
SELECT e.emp_ID, e.firstname, e.lastname, COUNT(q.QualID) AS
total_qualifications
FROM Employee e JOIN Qualification q ON e.emp_ID = q.Emp_ID
GROUP BY e.emp_ID, e.firstname, e.lastname
ORDER BY total_qualifications DESC;
```

JOINs Employee and Qualification tables. Since each employee has exactly 1 entry, all return count = 1.

Employees With At least One qualification

employees_with_qualification
60

Positions and Their Qualifications

Result Grid | Filter Rows:

Position	total_qualifications
HR Assistant	1
HR Manager	1
IT Support	1

Employees With Highest No of Qualifications

emp_ID	firstname	lastname	total_qualifications
1	John	Doe	1
2	Jane	Smith	1
3	Alice	Brown	1
4	Bob	Johnson	1
5	Charlie	Williams	1
6	Dave	Miller	1
7	Eve	Davis	1
8	Frank	Wilson	1
9	Grace	Moore	1
10	Harry	Taylor	1
11	Irene	Anderson	1
12	Jack	Thomas	1
13	Karen	Jackson	1

Leave & Absence Patterns

Q13: Which year had the most employees taking leaves?

```
SELECT YEAR(date) AS year, COUNT(DISTINCT emp_ID) AS
employees_on_leave
FROM Leaves GROUP BY YEAR(date) ORDER BY employees_on_leave DESC;
```

Uses YEAR() function to extract year from DATE field. All 60 leaves are in 2024.

Q14: Average number of leave days per employee per department?

```
SELECT jd.jobdept, AVG(emp_leave_count) AS avg_leaves
FROM (SELECT emp_ID, COUNT(*) AS emp_leave_count FROM Leaves GROUP
BY emp_ID) AS sub
JOIN Employee e ON sub.emp_ID = e.emp_ID
JOIN JobDepartment jd ON e.Job_ID = jd.Job_ID GROUP BY jd.jobdept;
```

Uses a SUBQUERY to count leaves per employee first, then JOINs to get department average.

Q15: Which employees have taken the most leaves?

```
SELECT e.emp_ID, e.firstname, e.lastname, COUNT(l.leave_ID) AS
total_leaves
FROM Employee e JOIN Leaves l ON e.emp_ID = l.emp_ID
GROUP BY e.emp_ID, e.firstname, e.lastname ORDER BY total_leaves
DESC;
```

Each employee has 1 leave record, so all return total_leaves = 1.

Most Employees Taking Leaves

year	employees_on_leave
2024	60

AVERAGE No Of Leaves Per Department

jobdept	avg_leaves
Operations	1.0000
Finance	1.0000
IT	1.0000
Marketing	1.0000
Engineering	1.0000
Sales	1.0000
HR	1.0000
Legal	1.0000

Most leaves Taken By which Employee

emp_ID	firstname	lastname	total_leaves
1	John	Doe	1
2	Jane	Smith	1
3	Alice	Brown	1
4	Bob	Johnson	1
5	Charlie	Williams	1
6	Dave	Miller	1
7	Eve	Davis	1
8	Frank	Wilson	1
9	Grace	Moore	1
10	Harry	Taylor	1

Leave & Absence Patterns

Q16: Total leave days company-wide

```
SELECT COUNT(*) AS total_leave_days FROM Leaves;
```

Result: 60 total leave records for 2024.

	total_leave_days
▶	60

Q17: Leave vs Payroll Correlation

```
SELECT e.emp_ID, COUNT(l.leave_ID) AS total_leaves,
       SUM(p.total_amount) AS total_salary
FROM Employee e
LEFT JOIN Leaves l ON e.emp_ID = l.emp_ID
LEFT JOIN Payroll p ON e.emp_ID = p.emp_ID
GROUP BY e.emp_ID;
```

Uses LEFT JOIN to retain all employees even if no leave. Correlates leave count with total salary.

	emp_ID	total_leaves	total_salary
▶	1	1	40500.00
	2	1	45500.00
	3	1	39500.00
	4	1	43500.00
	5	1	38500.00
	6	1	42500.00
	7	1	44500.00
	8	1	45500.00
	9	1	40500.00
	10	1	38500.00
	11	1	49500.00
	12	1	51500.00
	13	1	50500.00
	14	1	48500.00

Payroll & Compensation Analysis

Q18: What is the total monthly payroll processed?

```
SELECT YEAR(date) AS year, MONTH(date) AS month,
       SUM(total_amount) AS total_monthly_payroll
FROM Payroll GROUP BY YEAR(date), MONTH(date) ORDER BY
year, month;
```

Extracts year and month from the payroll date. Groups and sums payroll per month. Result: April 2024 total = 2,77,800.

year	month	total_monthly_payroll
2024	4	2778000.00

Q19: What is the average bonus given per department?

```
SELECT jd.jobdept, AVG(sb.bonus) AS avg_bonus
FROM SalaryBonus sb JOIN JobDepartment jd ON sb.Job_ID =
jd.Job_ID
GROUP BY jd.jobdept;
```

Averages the bonus column per department. Finance and Legal show highest average bonuses.

jobdept	avg_bonus
Operations	9687.500000
Finance	10666.666667
IT	10444.444444
Marketing	9125.000000
Engineering	12571.428571
Sales	11214.285714
HR	8171.428571
Legal	13300.000000

Payroll & Compensation Analysis

Q20: Which department receives the highest total bonuses?

SELECT jd.jobdept, SUM(sb.bonus) AS total_bonus
FROM SalaryBonus sb JOIN JobDepartment jd ON sb.Job_ID = jd.Job_ID
GROUP BY jd.jobdept ORDER BY total_bonus DESC LIMIT 1;

Result: Finance Department — highest total bonus allocation.

	jobdept	total_bonus
▶	Finance	96000.00

Q21: Average Adjusted Salary After Leaves

SELECT AVG(total_amount) AS avg_adjusted_salary FROM Payroll;

Averages the total_amount field from Payroll which already accounts for leave deductions.

	avg_adjusted_salary
▶	46300.000000

Challenges Faced

Technical Challenges

- Defining correct table relationships and ensuring accurate use of foreign keys across 6 tables
- Maintaining data consistency with cascading updates and deletes (ON DELETE CASCADE vs ON DELETE SET NULL)
- Writing complex multi-table JOINs for reports involving employee roles, leaves, and payroll simultaneously
- Ensuring all DATE fields follow YYYY-MM-DD format for reliable YEAR() and MONTH() based analysis
- Preventing duplicate entries using UNIQUE constraints especially on emp_email field

What I Learned

- Hands-on experience with ER diagram design and normalization
- Mastered JOIN types: INNER JOIN, LEFT JOIN across 3+ tables
- Used subqueries and window functions for advanced analytics
- Understood real-world HR data patterns — leave tracking, payroll processing
- Gained confidence in writing production-quality SQL queries

Conclusion & Key Findings

Key Findings

- **60 employees are distributed across 10 departments with Finance having the most job roles (9)**
- **Finance Director is the highest paying role at 170,000/month; total company payroll for April 2024 is 2,778,000**
- **All 60 employees have qualifications on record; roles like CTO and CFO carry the highest skill requirements**
- **All 60 leave records fall in 2024; every employee took exactly 1 leave during the period**
- **Finance department leads in both total salary allocation and total bonus payout**

Project Impact

This project demonstrates the ability to:

- **Design a normalized relational database from scratch**
- **Write 21 analytical SQL queries across 5 business domains**
- **Identify business insights from HR, payroll, and leave data**
- **Apply real-world data constraints: PK, FK, UNIQUE, CASCADE**

Tools Used

- **MySQL 8.0 | MySQL Workbench**
- **SQL: DDL, DML, DQL, Aggregations, JOINs, Subqueries**

THANK YOU